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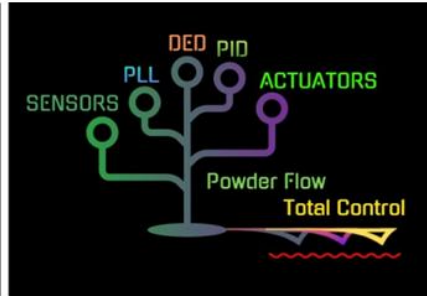
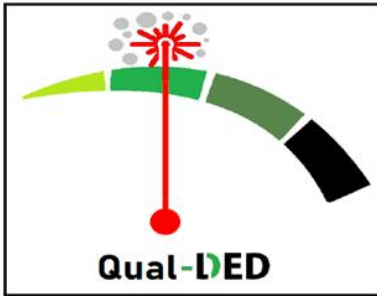
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TITOLO

The Problem of Powder Flow Control in DED Machines.

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The problem of powder flow control in DED machines

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There are many ways to determine with sufficient precision the mass flow rate of a homogeneous fluid flowing confined within a pipeline, but when the “fluid” is a heterogeneous mixture in different states of aggregation e.g. solid - gas, the problem can be difficult to solve.

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1. Introduction to the problem

The objective of this document is to realize analysis and offer solutions to the problem of in powder real control and regulation of the powder mass flow rate in Directed Energy Deposition (DED) machines, which use powder as raw material.

In DED machines, one of the main problems is the regulation of the powder, introduced through the laser head, which feeds the additive production process. The Figure 1 shows a schematic view of the powder feeder. The powder falls from the reservoir and is pushed by the stripper into a hose, which is connected with a supply of Argon. Argon is the vehicle of the powder to the laser head and it carries the powder through a tube with an internal diameter of 4mm to the outlet located in the laser head.

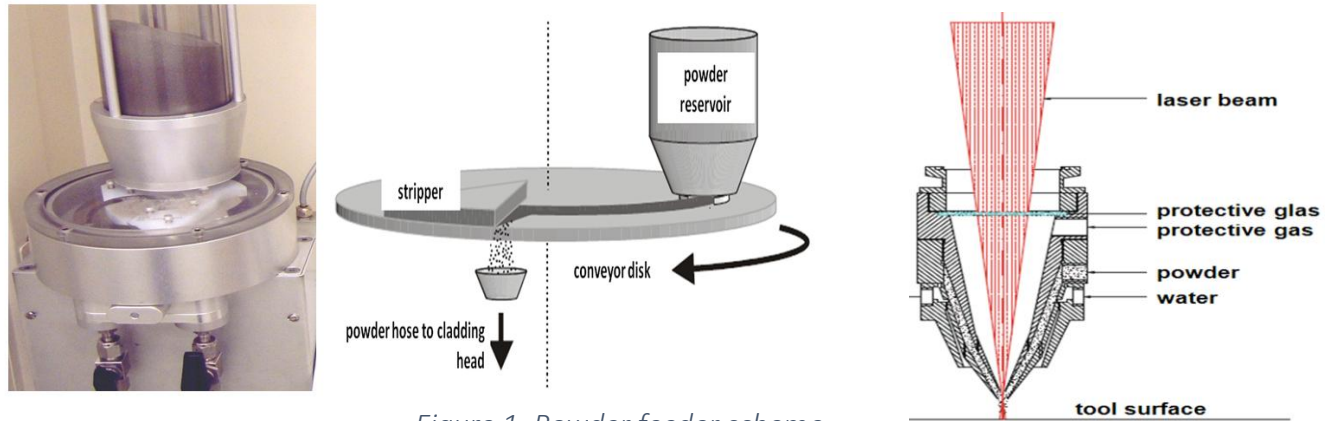


Figure 1. Powder feeder scheme.

The powder flow rate for most applications can be set between $2.5 \left[\frac{g}{min} \right] \div 50 \left[\frac{g}{min} \right]$. If converted into grams per second, a setting range of $0,083 \left[\frac{g}{s} \right] \div 0,83 \left[\frac{g}{s} \right]$ is obtained; such a small amount of powder is imperceptible compared to the sensitivity of a traditional powder feeder.

The main shortcomings of the powder feeder are the following:

- The quantity emitted by the powder feeder is irregular;
- There is no feedback controlling the flow rate;
- Even if there was a feedback, it is usually far from the laser head and the flow change would not be fast enough to eventually correct the flow rate;

As a result of this the powder flow is affected by large variations that can be **up to 30% of the total set powder flow**.

There are many ways to determine with sufficient precision the mass flow rate of a homogeneous fluid flowing confined within a pipeline, but when the “fluid” is a heterogeneous mixture in different states of aggregation e.g. solid - gas, the problem can be difficult to solve. In this document analyzes the most relevant physical principles to this end will be analyzed, the most promising selected and tested and the test results will serve to select one candidate to be coupled with the melt pool control concept (T3.3) so as to control the energy input per unit of mass in the laser material deposition process.

2. Theoretical analysis of the monitoring and control

2.1. Object of measurement

As exposed in the introduction, the main interest in this study is to measure the mass input of powder into a melt pool of a laser metal deposition process. However, this powder is brought into the melt pool by a stream of Argon. Although the fluid is thus composed of one gas and one solid fraction, it has a volumetric ratio of 99:1 in favor of Argon, which is why the mass is going to be treated as a fluid.

In general, when a fluid is placed in motion inside a pipe of known section, it has a measurable physical quantity, which varies with the speed of the fluid. This physical quantity is known as flow rate. There are two types of flow measurement:

1. **Volumetric flow:** The volume flow rate is defined in the following equation:

$$\dot{V} = Sv \left[\frac{m^3}{s} \right]$$

Where the symbol $\dot{V}$ represents the volume flow rate of a fluid flowing at a v velocity inside a S section pipe

2. **Mass flow:** The mass flow rate is defined in the following equation:

$$\dot{m} = \rho Sv \left[\frac{Kg}{s} \right]$$

Where the symbol $\dot{m}$ represents the mass flow rate of a fluid flowing at a speed v inside a pipe of section S with a known specific weight of ρ .

The difference between the two flow rates is, in fact, in considering (**mass flow rate**) or not considering (**volumetric flow rate**) the mass of the fluid. Consequently, a **volume flow rate** will give us a measure of how much volume of fluid will flow in the unit of time inside the tube, while a **mass flow rate** will give us a measure of how much of fluid will flow in the unit of time inside the tube.

In the case of powder flow in DED systems, the selected option is to measure the mass flow, for two reasons:

- 1) For the calculation of the energy density (J/mm³) it is not sufficient to know the designed height of the clad layer, but the real height which depends on the mass of the input powder; hence, in DED, the accounting of the powder in the process is always considered in gr/s or gr/min, which is why it is more adequate to measure directly the mass flow.
- 2) more important, since there is a small volume of powder in comparison to the Argon, but the mass of both integrands of the fluid are similar, given the specific masses of metals used in DED (steel, Inconel, aluminum, etc.) and Argon, on the other side.

2.2. Selection of the control concept

Before we introduce different monitoring concepts, we must discuss the control systems, since there is a clear dependence of the selected monitoring concepts on the type of the control we want to make in the DED process.

In general, there are two main types of control systems. The first, called **Open-Loop Control Systems**, are used when the system does not require much precision, is slow, cost-efficient and usually the control is performed by a specialized operator, who takes responsibility for being at the same time controller, sensor and actuator of the system under control.

The second type of control is called **Closed-Loop Control Systems**- these systems are used whenever it is necessary to have a very precise, robust and fast process control, but to build a similar control system it is necessary to *constantly* measure the physical quantity we need to control. The faster we measure and acquire the value of the physical quantity we want to control, the more accurate the control will be.

To be able to control the energy input per unit of mass, both energy and mass input, that is, the flow of powder require appropriate precision, since even the smallest deviation affects the accumulation of thermal tensions and the loss of geometrical control over the manufactured part. Hence, it is necessary to be able to measure the powder flow in short time lapses, in order to make the necessary corrections during the additive manufacturing process. Also, the powder flow rates at stake are really small, so not only the accuracy is needed, but also it is necessary to use a high *sensitivity* sensor.

Therefore closed-loop control system of small quantities of powder require the following sensing principle characteristics:

- High sensitivity (due to small amount of powder)
- High speed to measure and acquire data (for accurate control)

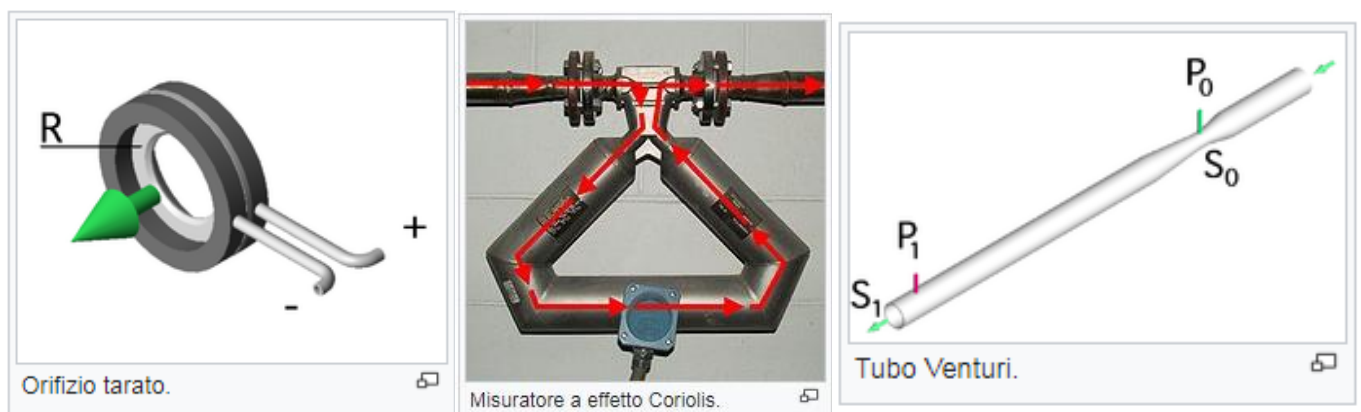
2.3. Analysis of monitoring concepts

2.3.1. Measurement methods

We can make an initial distinction between the measurement methods, used to measure fluid's flow rate, into:

- mechanical methods,
- electrical methods,
- magnetic methods
- acoustic methods
- optical methods.

2.3.1.1. Mechanical methods



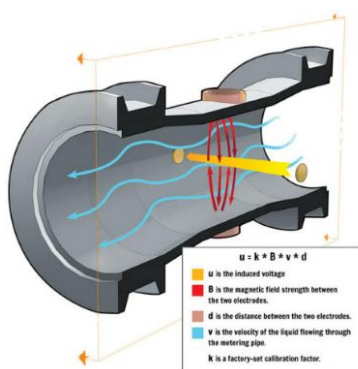
These were the first methods of measuring the flow rate of homogeneous or near-homogeneous fluids to appear in the engineering. Their function is based on the notions of classical mechanics. The measurement is direct or almost direct, but it is applicable to quasi-static fluids or fluids with not too high

velocities, such as the speed of water flowing in a water pipe, which makes it difficult to apply in the case of powder flow in DED.

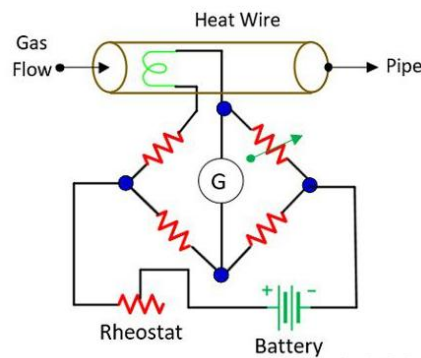
Among the systems that use the mechanical methods, the most relevant are:

- Variable area flow meters
- Calibrated orifices
- Venturi tube
- Pitot tube
- Coriolis effect meter
- Volumetric meters
- Turbine meters

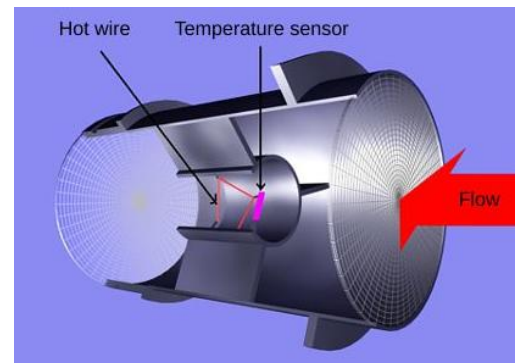
2.3.1.2. Electrical methods



Magnetic Meter



Hot Wire Anemometer



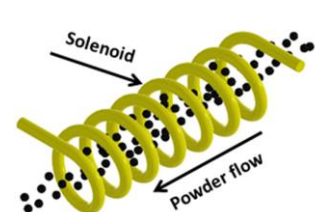
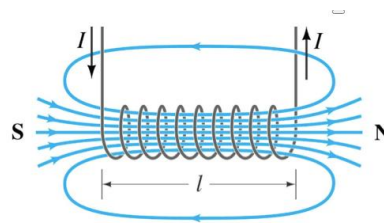
These methods were introduced thanks to the discovery of electricity. They work upon the principle of electrical quantities measurement, having the advantage of *very high frequency response*, compared to the slowness of traditional purely mechanical meters.

These methods are based on measurements of current and voltages, for example in **hot wire anemometers** a wire of known resistance is used, in which an electrical current is run, which then increases the temperature of the wire itself, as a result of which the electrical resistance of the wire will vary further when cooled by a flux e.g. gaseous. As it is known, voltage and current resistance are linked by OHM's law ($V=R \cdot I$) so with a Wien bridge electric circuit it is possible to trace the flow rate of the fluid according to the variation of the voltage value produced by the variation of the resistance of the hot wire. Also in **Magnetic meters** the velocity of the fluid is estimated based on the variation of electrical voltage, but in this case an alternating magnetic field is applied perpendicular to the direction of the fluid, which must be conductive, two electrodes in contact with the fluid record the variation of voltage only if there is relative movement between the fluid and the applied magnetic field.

Among the systems, which use the electrical methods, the most relevant are:

- Magnetic Meters
- Hot wire anemometer
- Hot wire meter

2.3.1.3. Magnetic method



Magnetic inductance is a measure of the tendency of an electrical conductor to oppose a change in the electric current flowing through it when an external factor (or the same current through the wire) is altering the magnetic field. Hence, any change in the magnet field strength, caused by a change in current, induces an electromotive force (EMF) in the conductor that opposes the voltage creating the change of current. This physical effect is known as electromagnetic induction and is described by the laws of Faraday and Lenz. This induced voltage in the circuit is called back Electro Magnetic Force (EMF), to reflect its opposing character. Therefore, the magnetic inductance is defined as the product of the ratio of the induced voltage to the rate of change of current causing it and a proportionality factor that depends on the geometry of circuit conductors and the magnetic permeability of nearby materials. For a specific case of an induction coil (solenoid), it can be expressed as:

$$L = \mu \frac{N^2 S}{\ell}$$

where the L is the inductance, expressed in Henry's (H), μ is the magnetic permeability of the material inside the coil, N is the number of the coil turns, S is the section area of the coil and ℓ is the coil length. Hence for a fixed induction coil, in which N , S and ℓ are constant, the value of the inductance will be directly proportional to the magnetic permeability of the material that resides in the interior of the coil at the moment of measurement. This could be measured by an electronic setup as shown on the following figure.

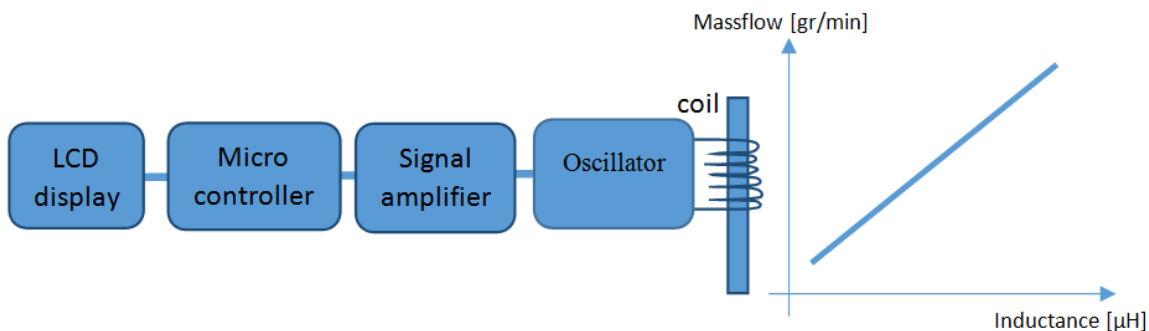
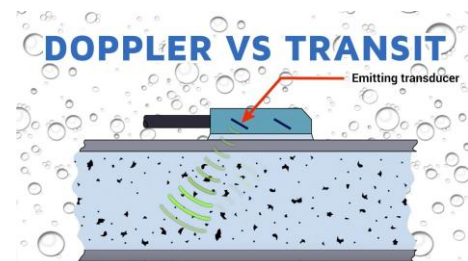
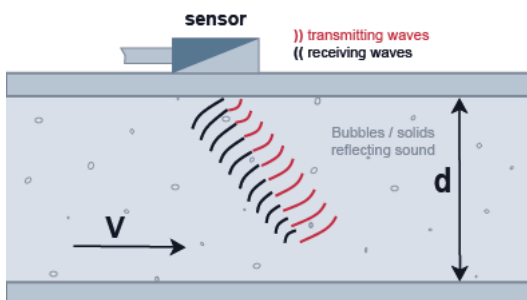


Figure 1. Simplified scheme of Inductance meter.

2.3.1.4. Acoustic methods



Acoustic methods are based on the difference in the speed of sound propagation in the different materials. In the following table we can see the speed of sound propagation in some materials:

Speed formula $\longrightarrow v = \frac{s}{t} \left[\frac{m}{s} \right]$

Where:

- v : is the speed of the sound wave when the fluid is static
- s : is the space traveled by the sound wave
- t : is the time spent by the wave to cross space s

Material	Density(g/cm)	Speed(m/s)
Copper	8.90	6420
Steel	7.86	5940
Beryllium	1.93	12890
Aluminium	2.58	6420
Water	1.00	1496
Ethanol	0.79	1207
Air	0.00139	331.45
Helium	0.000178	965
Fat	0.95	1450
Muscle	1.07	1580
Skull bone	1.91	4080

In general, knowing the sound wave speed for each one material is possible to calculate the crossing time for a sound wave that travels to certain space filled of known material. In accordance with the speed formula as distance over time, if we assume that v and s are constants, the crossing time t have to be also constant, but if the fluid is in movement the crossing time have to change because of the relative movement between sound wave source and the fluid. It is therefore possible determine the relative speed v' of the fluid in movement

$$v' = v_f - v$$

Where:

- v : is the speed of the sound wave when the fluid is static
- v_f : is the speed of the sound wave in the moving fluid
- v' : is the relative speed between the sound source of the sound wave and the fluid

There are two main types of ultrasonic flowmeters: Doppler and transit time. While they both utilize ultrasound to make measurements and can be non-invasive (measure flow from outside the tube, pipe or vessel), they measure flow by very different methods.

1. Ultrasonic transit time flow meter

The measurement principle in this instrument is based on the recognition of the period of transit of ultrasonic pulses between two sensors (A and B). The time transit of the transmitted signal in the direction of scrolling of the fluid is shorter than the fluid transmitted against the direction of flow. The difference between the signals ($t_2 - t_1$) is therefore proportional to the average flow velocity V_m . The concept is shown at the Figure 2.

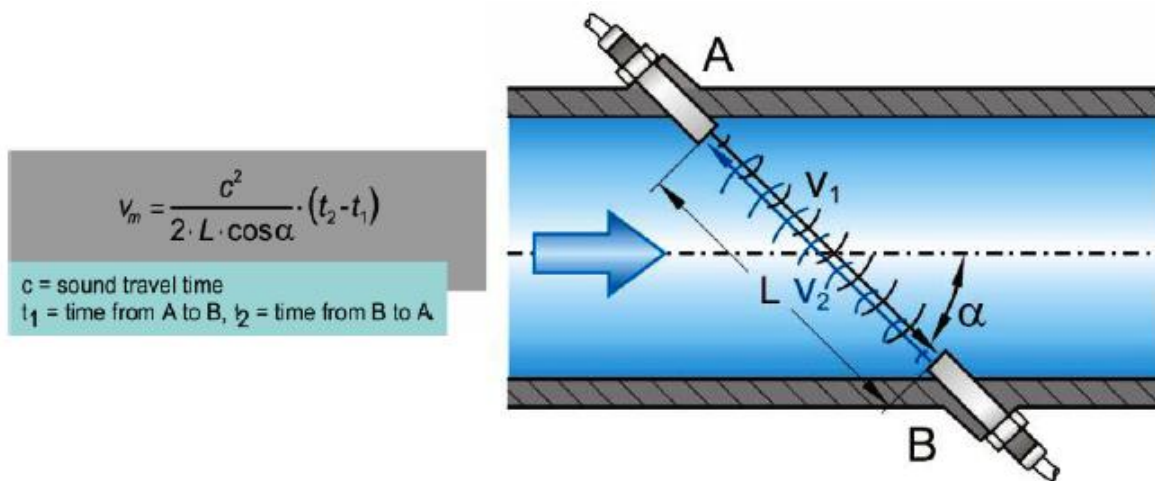


Figure 2. Ultrasonic transit time flow meter.

The transit time of the ultrasonic pulses is determined in precise way using the cross-correlation method. This measurement method has the main advantage that it is significantly less sensitive to interference than in other common measurement methods. However, this method works when we are dealing with a homogeneous fluid that fills the entire measurement volume between the two sensors. If the presence of solid particles in suspension or gas bubbles is not negligible, then the concept doesn't have good performance.

2. Ultrasonic Doppler flowmeters

The Doppler effect concept of measuring the flow is based on a mathematical relationship that links the velocity of the material flow with the variation of the wavelength of the sound wave, being this relationship known as the Doppler effect.

$$f' = \frac{v}{v \pm v_{s,r}} f$$

Where:

f : Frequency transmitted by the sound source

f' : Frequency received by the observer

v : Speed of sound waves in the material

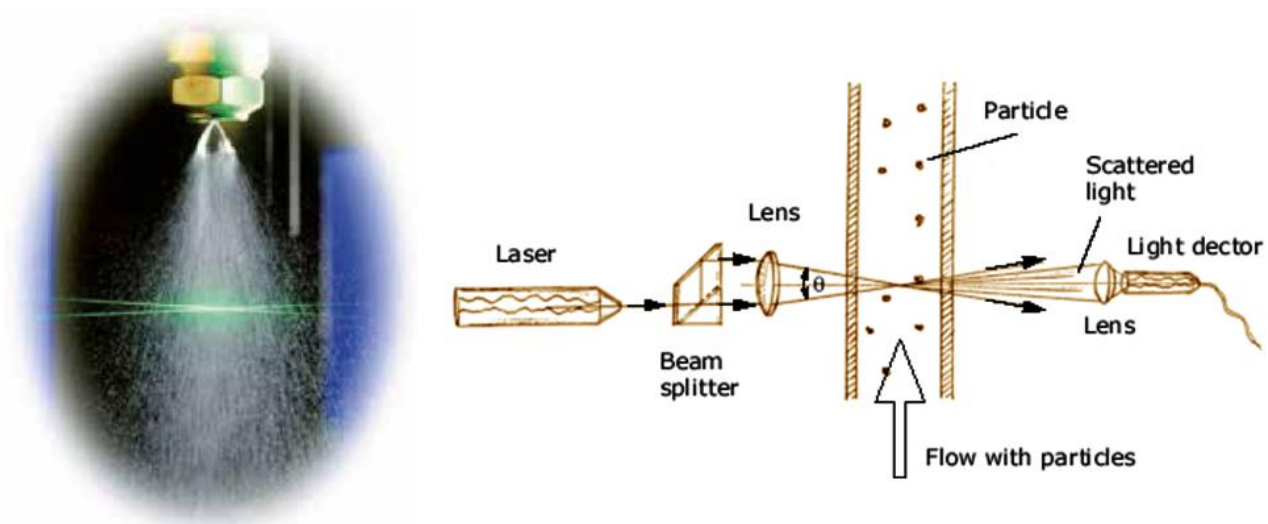
$v_{s,r}$: Speed between wave sound source and the observer (considering only the component in the direction that unites the source and the observer)

In this case, the transducer emits a train of ultrasounds which is rebounded by the particles returning to the emitter but with a different frequency depending on the speed of the measured fluid and therefore knowing the area returns to the volume flow.

This method is used in the following measuring instruments:

- Ultrasonic meters
- Doppler effect ultrasonic meters

2.3.1.5. Optical methods



The optical methods are based on the light change when it collides with a matter (particle, object, wall, etc.). The main relevant properties are wavelength and light power. The optical methods are the fastest detection method and are used in ultra-fast sensors. The main components used in optical measurement systems are the emitter of light, commonly **laser**, and the receiver of the light modified by the media through which the light travels, where the most common are **optical avalanche sensors**.

Laser sources:

LASER, which stands for Light Amplification by Stimulated Emission of Radiation, is an optoelectronic device able to emit a coherent beam of light or an electromagnetic wave that maintains a certain phase relation during its propagation. The laser is essentially composed of 3 parts:

- ✓ **An active medium**, i.e. a material (gas, crystal, liquid) that emits light;
- ✓ **A pumping system**, which supplies energy to the active medium;
- ✓ **An optical cavity**, or optical resonator, i.e. a light trap.

The laser uses the **active medium** to emit electromagnetic radiation (**photons**) when activated. The wavelength of the emission depends on the active medium. The active medium can be gaseous (e.g. carbon dioxide, helium and neon mixture, etc.), liquid (solvents such as methanol, ethanol or ethylene glycol, to which chemical dyes such as Coumarin, Rhodamine and Fluorescein are added) or solid (ruby, neodymium, semiconductors, etc.). The **pumping system** supplies energy to the active medium, leading it to excitation with photon emission. It is based on optical pumping (laser diode strobe lamps, etc.), electronic shock (electrical discharge in gas with direct current source, pulsed, radio frequency, or a combination thereof), etc. The transmitted radiation is normally concentrated through an **optical cavity** with internal reflecting walls and a semi-reflective exit area. This last surface is the only one that allows the beam to escape, which is then processed and repositioned through a series of lenses and mirrors to ensure that the resulting beam has the desired position, concentration and amplitude.

Lasers have revolutionized the way to measure practically all physical quantities both tangible and intangible. The advantage lies at the *very high frequency response* being suitable for very high precision measurements even if the moving fluid has a relatively high speed compared to the measuring system. They are also insensitive to electromagnetic fields, they are not invasive, i.e. they do not interact in any way with the fluid under measurement and, with appropriate software processing techniques, it is possible to obtain in real time all the physical quantities related to the fluid under examination such as for example: mass flow rate, concentration, density, speed, temperature, volume, etc.

Measurement techniques:

Different variation of laser properties are used for measuring different parameters:

1. **light scattering.** If we make a laser beam interact with a very small particle (even a very small particle of the micro or nanometer order), we obtain the dispersion of some photons in random directions, being known this phenomenon as light scattering. Because of this, the laser beam received after interaction with the particles will have a lower light intensity.

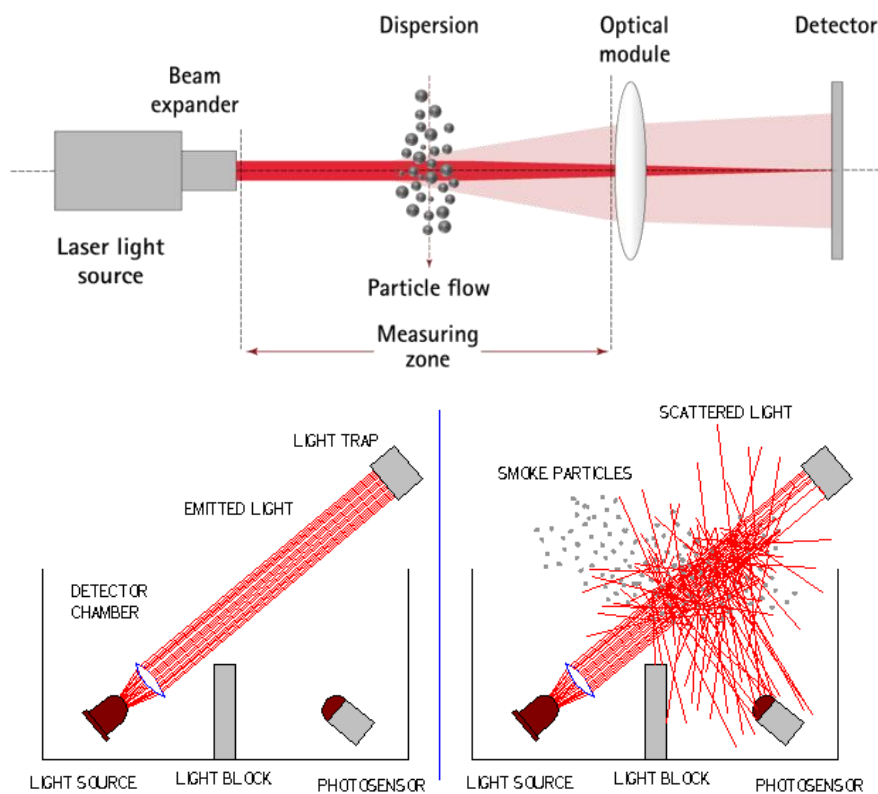


Figure 3. Light scattering as a measurement concept: dispersion (up) and light trap (down)

2. **Light Absorption:** all materials absorb light; thanks to this principle, it has been possible to realize a very important measuring instrument called Mass Spectrophotometer.

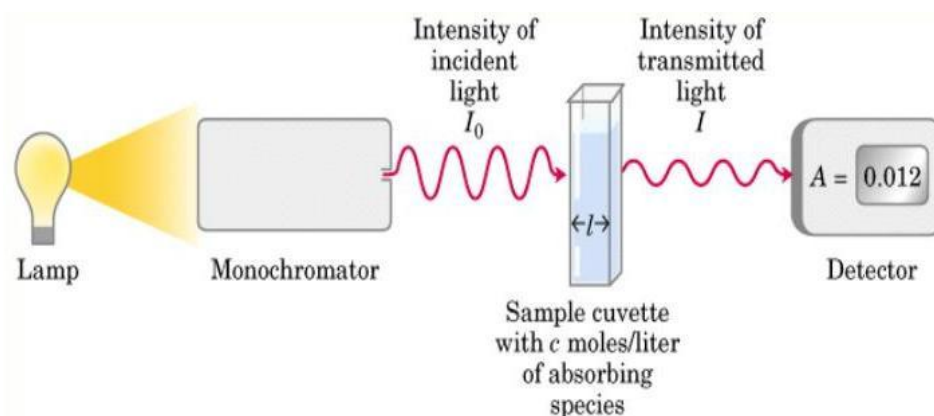
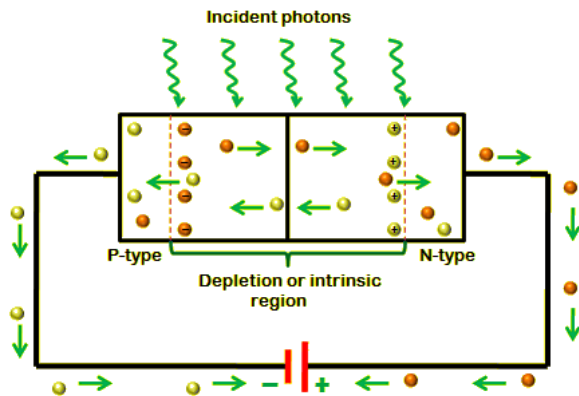


Figure 4. Light absorption as a measurement concept.

3. **Spectroscopic techniques:** spectroscopic techniques are all those techniques based on the interaction between matter and electromagnetic radiation. Light, heat and other electromagnetic radiation are vibrations of magnetic and electric fields that propagate in space. Hence, the principle consists in atoms or molecules, which, being in energy fields, can absorb defined quantities and energy characteristics, passing to higher (absorption) or lower (emission) energetic states.

Optical sensors:

In any of the above-mentioned cases, the indicative value for the measurement of the mass flow of the particles is the **variation of electric signal as the laser interacts with the interaction of powder particles**. There are different, more or less complex detection concepts, but the most adequate for this purpose is a photodiode sensor, which is why we are going to focus on it.



Avalanche Photodiode

What is it ?
How it works ?



An avalanche photodiode is a device that transform optical signal into electrical signal. The photodiode can be operated in one of two modes:

1. Photoconductive (reverse-bias)

in this configuration the photo diode works as if it were a variable resistor, but the response to incident light is not linear, in fact these photodiodes are used to obtain an amplification of a weak light signal received at a given wavelength, in this case you can get high sensitivity with regard to the signal received and then transduced.

2. Photovoltaic (zero-bias)

In solid state physics the photovoltaic effect is the physical phenomenon that occurs when an electron present in the valence band of a material (usually semiconductor) passes to the conduction band due to the absorption of a sufficiently energetic photon incident on the material. As is known, this effect is used in inpowderry for the direct production of electricity from the sun.

Since for our application we will need a very sensitive receiver sensor we will focus on the first type of photodiodes:

This photodiode must be sensitive to the wavelength emitted by the laser source. Therefore, the selection of the photo diode is done according to the wavelength of the selected laser source. It is frequent to use laser diodes in the visible spectrum, with wavelengths between **400 - 800 nanometers**, when one could use the photodiode **BPW34**, which is sensitive to all radiation between **400 – 1000nm**. *In any case, the sensor is selected according to the type and wavelength of the laser source.* In this example, having an efficiency of 80% on the wavelength of 670 - 680 nanometers the selection of the laser source should be done in that range of the wavelength, because we can get a high intensity laser beam on its surface, without running the risk of saturating it.

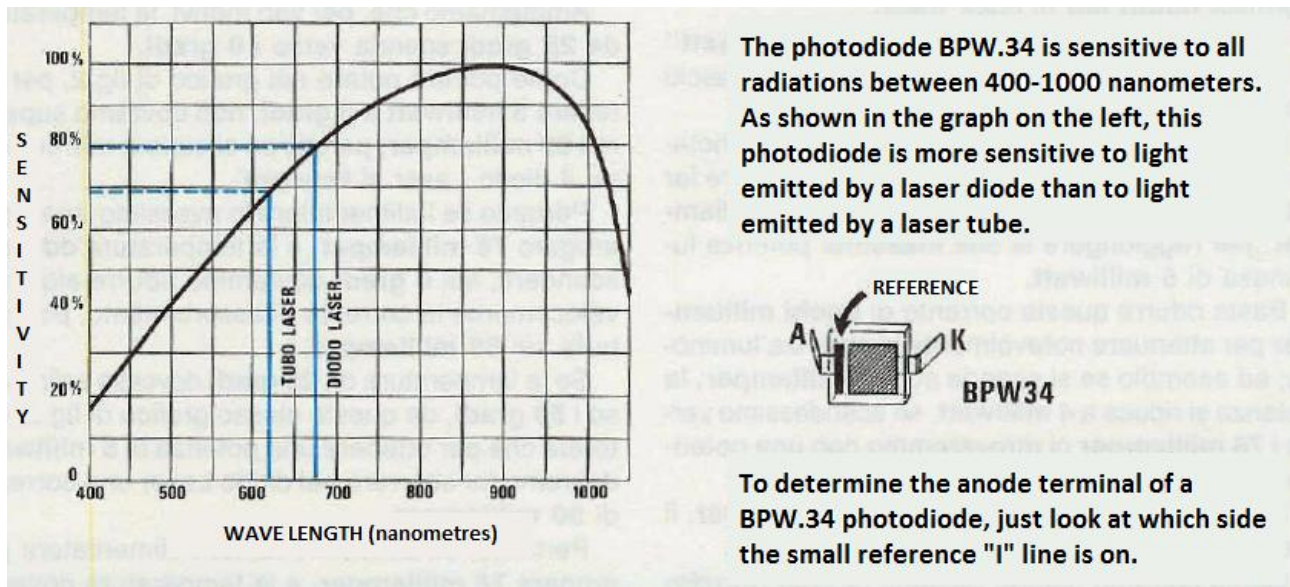


Figure 5. The relation of the sensitivity and laser source wavelength.

Apart from the wavelength, more parameters of laser are important to select the photo diode: **power** (commonly in order of mW), **physical size of the laser source**, **spot size** (from a few μm to a few mm), **spot shape** (which does not necessarily have to be a point), etc. Nowadays, considering especially the small size feature, the only type of Laser that has all the characteristics described above is the **diode laser**.

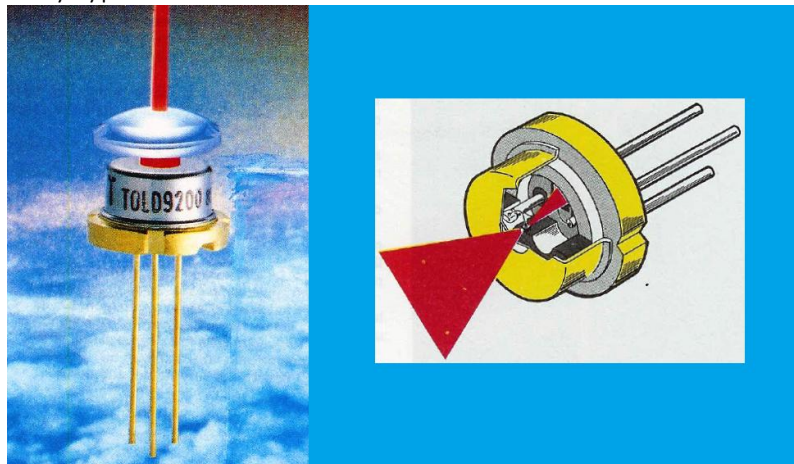


Figure 6. Diode-based laser source.

Like any other semiconductor, laser diodes are classified according to their power and wavelength, which is normally specified in nanometers. Currently, small laser diodes are divided into three categories:

- **GaAlAs Laser** - These diodes emit radiation over a wavelength of **750-880 nanometers**, non-visible for the human eye.
- **InGaAsP Laser** - These diodes emit radiation over a wavelength of **1300 - 1500 nanometers**, so like the previous ones are not visible to the human eye.
- **InGaAlP Laser** - These diodes emit radiation over a wavelength of **670-680 nanometers**. The light emitted by these laser diodes is visible to the human eye with a **red color**.

The laser diode emits an oval light beam with two different divergences:

- **Divergence parallel** to the joint; it is between 6 and 12 degrees,
- **Divergence perpendicular** to the joint; it is between 20 and 40 degrees

In practice, the beam of the laser diode widens in a fan pattern so that at a distance of a few centimeters, it will cover a considerably large area.

As a result, a laser diode without an appropriate lens will never be able to provide us with that small bright spot like any other type of



Figure 7. Diode-based laser source: propagation.

For an adequate detection, it is important to analyze the **brightness** of the laser source. The brightness perceived by the human eye is not a suitable instrument for the estimation of light power because, as is known, the perception of the various wavelengths by the human eye is not linear. Consequently, laser diodes of identical power but with different wavelengths will be perceived differently by the human eye.

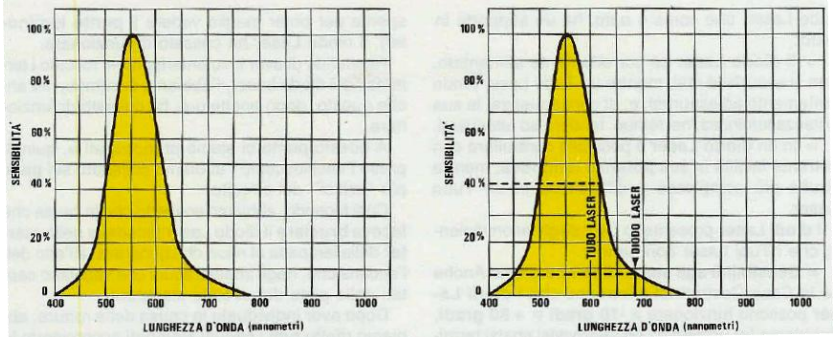


Figure 8. Laser source brightness.

In what refers to the **power** of the laser source, the main factor is its stability. This depends on the *diode type* used and the *control of the power* since, to have a precision instrument that measures a very low variation of the light emission, it is necessary to have a perfect stability of the main light emission. The next figure shows on of the examples of the laser diode that can be used in this application.

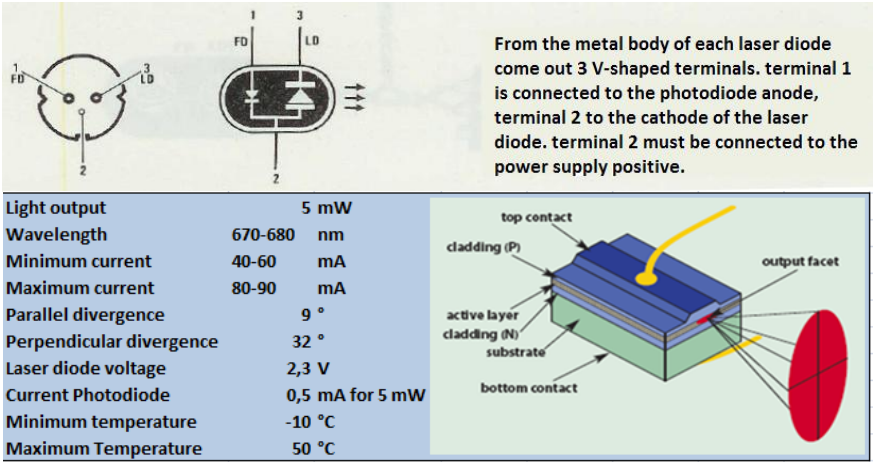


Figure 9. Diode laser source indicated for the powder flow measurement application (Laser Diode type HL.6711/G o TOLD.9201/STR).

To control the light emission, of all Laser Diode, a photodiode is inserted into the cavity where the light is produced. This photodiode is used to measure the light produced by the laser and provide the electrical control signal **to close the loop of a control system**. This ensures the stability of the output power avoiding fluctuations of light power that can be adjusted.

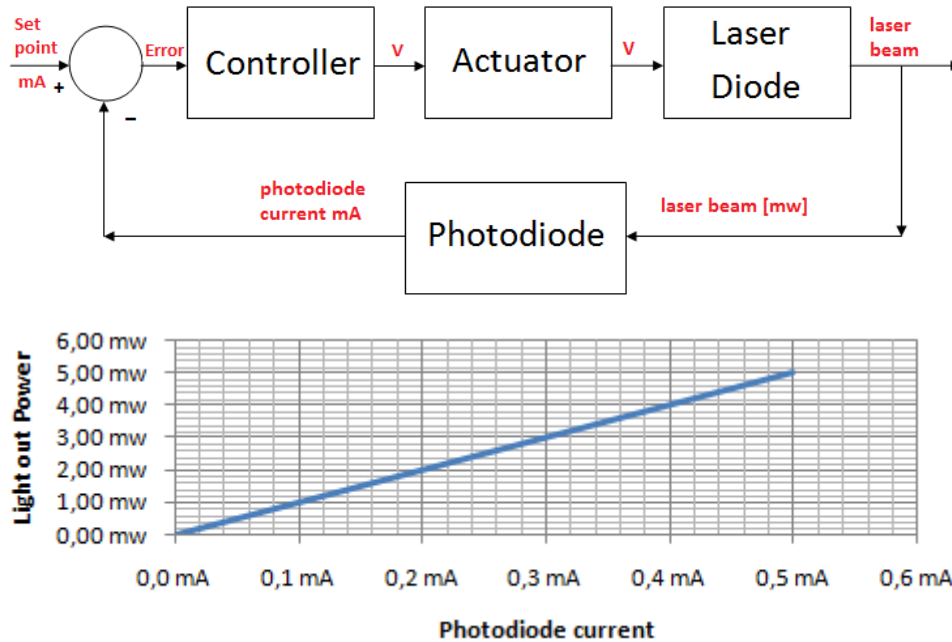


Figure 10. Scheme of a closed loop control of the laser source power (up), stabilized relation between the current and power.

2.3.2. Selection of the measurement methods for the testing

In light of what has been learned from the various flow measurement systems, and considering that in DED machines the working fluid is a heterogeneous mix of metallic powder is gas in very small quantities the most appropriate choice can fall on only two of the methods mentioned above:

- **Magnetic method:** This method as previously stated uses as a sensor a solenoid in which the workflow is made to flow inside, the variation of the powder mass flow will be related to the variation of the self-induction coefficient of the solenoid.
- **Optical method:** We will use this method considering the fact that even if the quantity of material circulating inside the machine is very small, the laser-photodiode system is still able to detect the very small variations in the flow thanks to the fact that every material, however small, absorbs, reflects and transmits light.

3 Monitoring concepts testing

3.1 Material and methods

For this experimental testing, the following existing equipment was used:

- ✓ Powder feeder GTV with two types of materials: stainless steel 316L and nickel-based alloy Inconel 625, granulometry of 45-100 μ m in both cases, varying the mass flow of the powder in the ranges 6.5 – 20 gr/min and 7.5-22.5 gr/min respectively. The argon flow dragging the powder was of constant flow (5l/min). The powder was let to flow in constant regime, taking the measurement after approximately 2 minutes.
- ✓ Robot arm ABB IRB 4600 25 kg load capacity

- ✓ Laser Head 3-6 Kw Nd:YAG, is a solid state laser that uses a neodymium-doped yttrium aluminum garnet (YAG) crystal as an active laser medium (Nd:Y3Al5O12).
- ✓ Electronic circuits made on purpose for each testing, according to the schemes and specifications, which will be exposed in each subsection where the particular test results are explained.
- ✓ Additional apparatus such as an oscilloscope, a radio receiver, etc. which were used for measurement of inductance and frequency.

3.2 Testing

3.2.1 Magnetic methods for powder flow measurement

Two principles for the magnetic-based powder flow measurement were performed: *magnetic inductance measurement* and *super heterodyne stabilized frequency measurement*.

3.2.1.1 Magnetic inductance measurement

The basis for the first testing is an important relationship is between the inductance and the resonance frequency (f_0). When the change of inductance is produced, an oscillator can register the frequency change that is directly related according to the diagram on the following Figure, which also show the scheme of the electronic setup needed. The advantage of this approach is that at the small inductance variation is small produce high frequency variation, so measuring frequency instead of inductance might be easier.

$$f_0 = \frac{1}{2\pi\sqrt{L_x C}} \quad \Rightarrow \quad L_x = \frac{1}{C(2\pi f_0)^2}$$

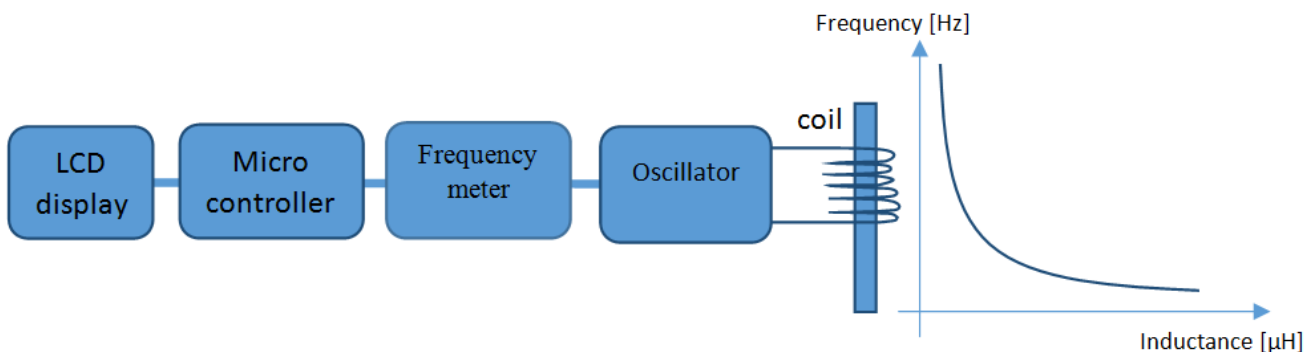


Figure 1. Simplified scheme of frequency meter.

3.2.1.2 Super heterodyne stabilized frequency measurement

Finally, another principle that can be used is the principle of super heterodyne receiver, contained in every radio receiver. The principle of work is the one used in radio emissions – a heterodyned (combined) signal is sent by the radio transmitter, consisting of the decided emission frequency (e.g. 100 MHz) and the voice or music converted into a 20-20.000Hz frequency. When the radio receives the signal (e.g. 100.000.020 Hz – 100.020.000 Hz), a local oscillator changes the incoming radio frequency signal to a higher or lower, fixed, intermediate frequency (IF). In other words, its “rests” the emission frequency (100MHz) and the demodulator extracts the audio or other modulation from the remaining IF radio

frequency; the extracted signal is then amplified by the audio amplifier and heard through the speakers. This can be done no matter what the original noise source is: a voice, music or even the frequency change induced by the change of magnetic inductance. In other words, if the change of frequency within the induction coil is transmitted to a radio receiver, if the frequency change is between 20 and 20.000Hz, it can be heard on the radio receiver. The following Figure shows the simplified scheme.

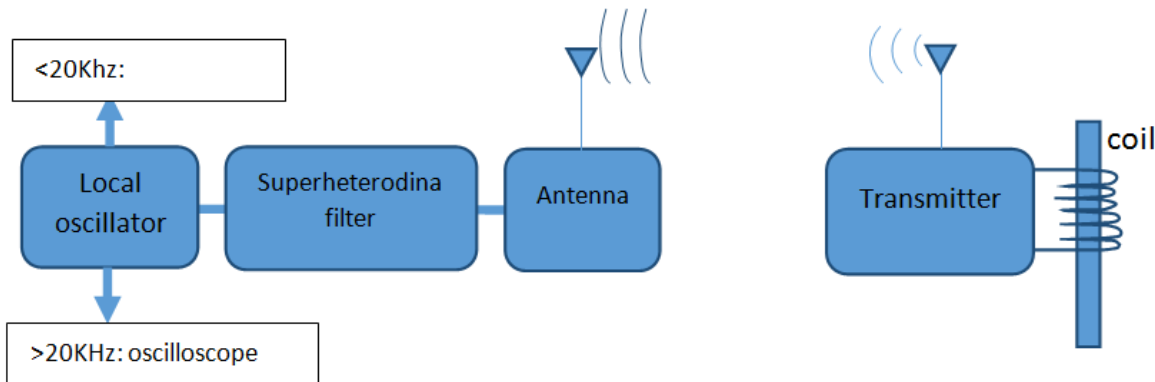


Figure 2. Simplified scheme of Rx - Transmission.

3.2.1.3 Results and discussion

The first set of measurements was taken using the inductance meter setup (Figure 1). At the following images, the experimental setup can be found.

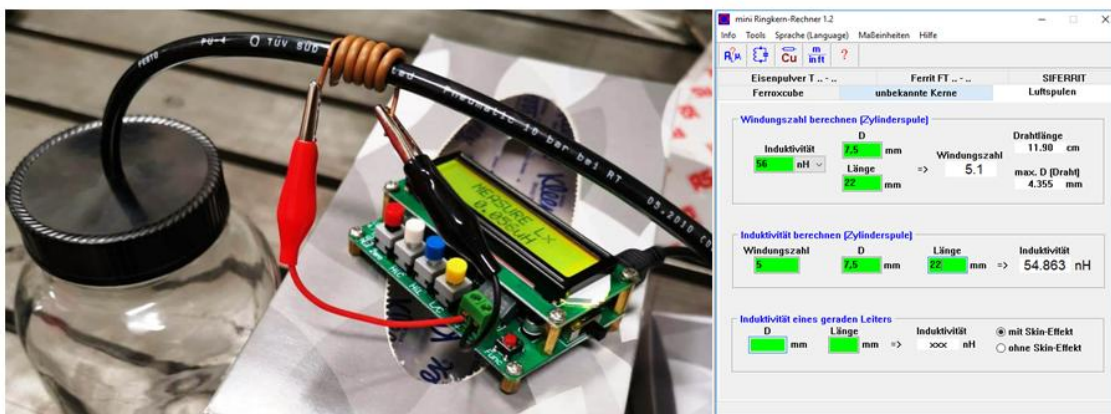


Figure 1. Experimental setup with the inductance meter.

The powder flow was conducted through a 6mm tube that, instead into the cladding nozzle, was discharged into a glass bottle. The coil had an approximate diameter of 7.5mm (centreline considered), 5 turns and an approximate length of 22mm. The resulting inductance was in agreement with the one theoretically calculated using the Eq1 (also backed up by an open source software called “mini Ringkern-Rechner” specially programmed for inductor design). However, the measurements showed that the inductance L has very low values according to the following table.

Testing material: steel	Mass within the measuring volume [gr]	ΔL [nH]
Offline testing		
Full dense steel	647,4	100
Packed powder (50-52%)	323,7	50
Continuous flow 15gr/min	0,25	0,0039

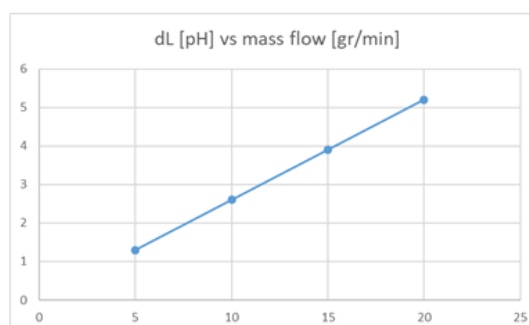


Table 1. Testing results (left) and extrapolated relation between the inductance change and the mass flow of powder.

Calculated over a unit of time (e.g. 1s), the volume of the mixture Argon-powder is 0.083 dm^3 , while the amount of powder contained in it is only 0.25gr, which may explain the small variation in the inductance caused by the 15gr/min of powder. In other words, the change of inductance introduced by the powder flow pass was very small and the difference between different powder flow rates were impossible to detect with a simple and cheap LCR meter employed in this testing.

Therefore, the selected method for the second set of powder flow measurement was to use the frequency change instead of the inductance (according to the Figure 2). The new circuit and the experimental setup are shown on the following Figure.

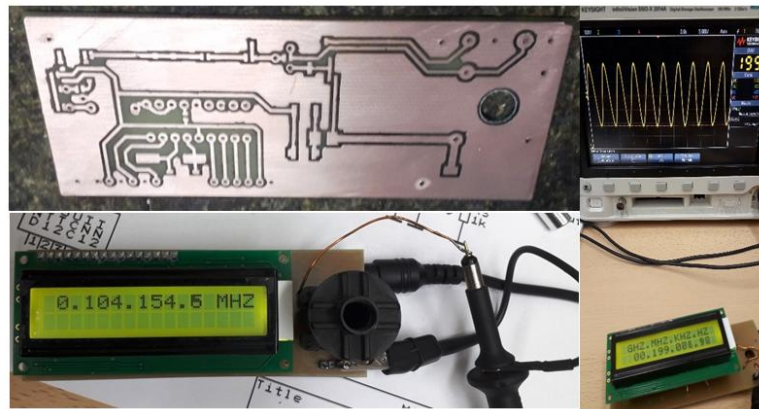


Figure 2. Experimental setup with the frequency meter.

Although the variation that was generated by the powder flow in this case was corresponding to a range of 2KHz, it was noted that the problem with measuring the frequency is the lack of stability in the measured value. *The oscillation of the frequency value was significant, probably because of the intermittent pass of the powder particles.* In other words, for low inductance values, the fluctuations were impossible to perceive and measure, the corresponding frequency change is higher, the fluctuations were higher as well, making difficult to measure a stable value. In summary, the value of the change was in the order of 2KHz, but the fluctuations had order of magnitude of 200-300Hz, without the stabilization and without a possibility to detect any change when the powder flow was modified from 5 to 20gr/min. What remains unclear if all the fluctuations come from the variable powder particle income or there is an influence of other factors as well (other magnetic fields that interfere with the signal).

For this reason, it was assumed to test the third principle – super heterodyne receiver. The following figure shows the setup for this test. Here an additional measure was undertaken: the circuit was placed into a metal box, emulating the Faraday's cage, to avoid other influences. The signal was received by the radio to be emitted on the speakers, but also it was connected to the oscilloscope to register the signal visually as well.

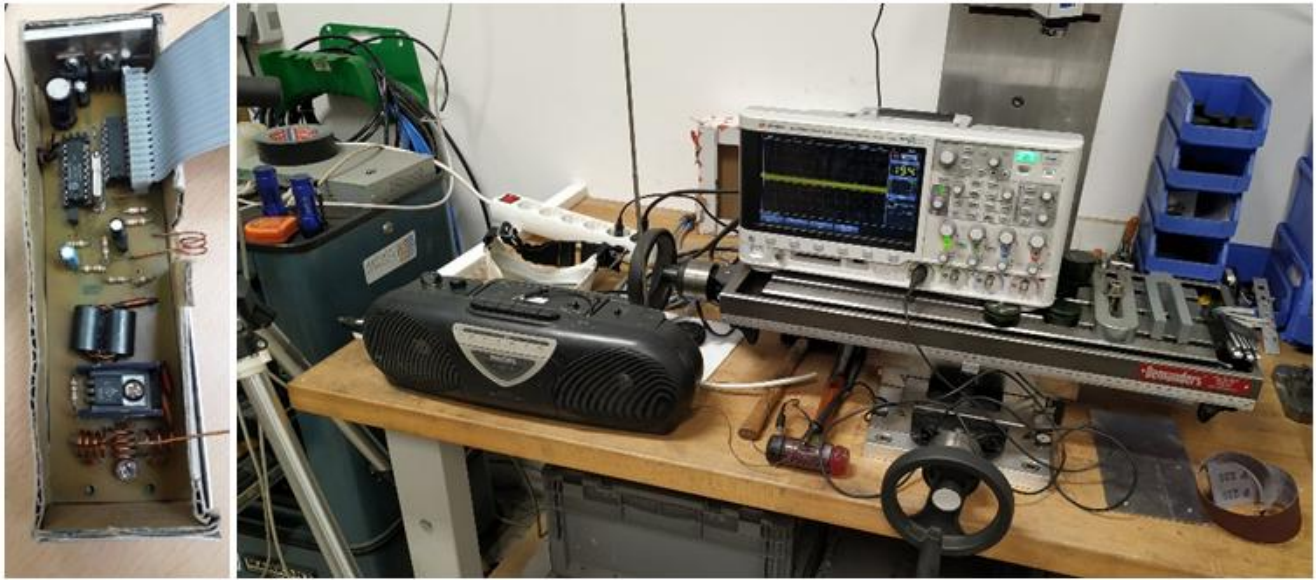


Figure 3. Experimental setup with the frequency transmission (first attempt).

The results were not conclusive either. There was a significant difference between the *on* and *off* status of the powder feeder, but there were no significant changes when the powder flow rate was changed.

3.2.2 Magneto-acoustic method

This acquisition method differs from the known methods because it correlates the variation of the solenoid self-induction coefficient (magnetic method) with the production of an audible acoustic signal (between 0-22 kHz) based on the variation of the powder flux, so it has been called **Magnetic-Acoustic method**. Given that the testing described in the “*Results and discussion*” section showed some shortcomings, a change of the measurement technique was introduced, In fact the final circuit is equipped by PLL phase locked loop circuit that ensure high stability of the frequency generated and transmitted to the super heterodyne circuit that is performed by a professional SMD radio circuit directly connected to the output of the PLL transmitter. The principle is similar to the first attempt but in this case there is no transmission through the air because the signal generated to the transmitter passes directly through the super heterodyne module, avoiding spurious magnetic field, and the transmitter is equipped by PLL which ensures a perfect stability of the base frequency in this case (87.7 MHz). The following Figure shows the simplified scheme.

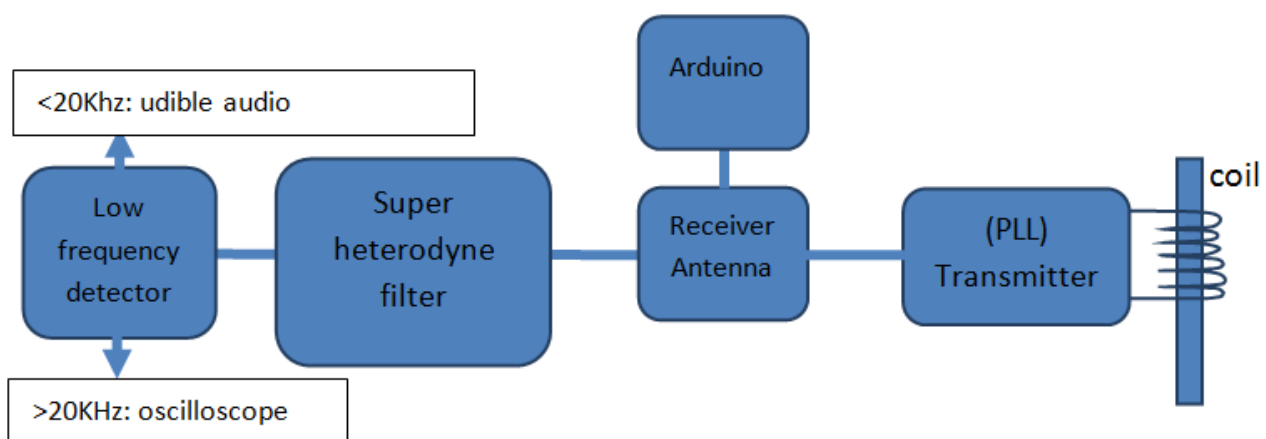


Figure 1. Simplified scheme of Rx – PLL transmission Arduino controlled

Unfortunately, because covid-19 emergency instead of using the Powder feeder GTV we have built a prototype of new powder feeder to make experiments not in situ. The prototype doesn't need gas to flow the powder, but the flowing is performed by vibrations of one metal sheet element that causing the powder to fall down through the PWD sensor. The first tests were carried out by introducing a variable quantity of 0-5 g of stainless-steel powder into the system.

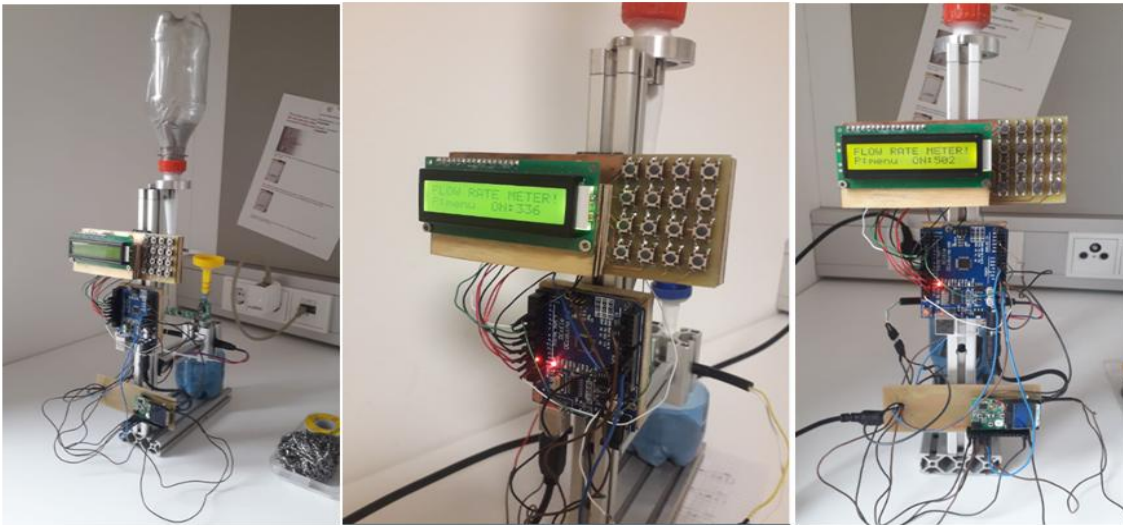


Figure 2. Experimental setup with the prototype of new powder feeder

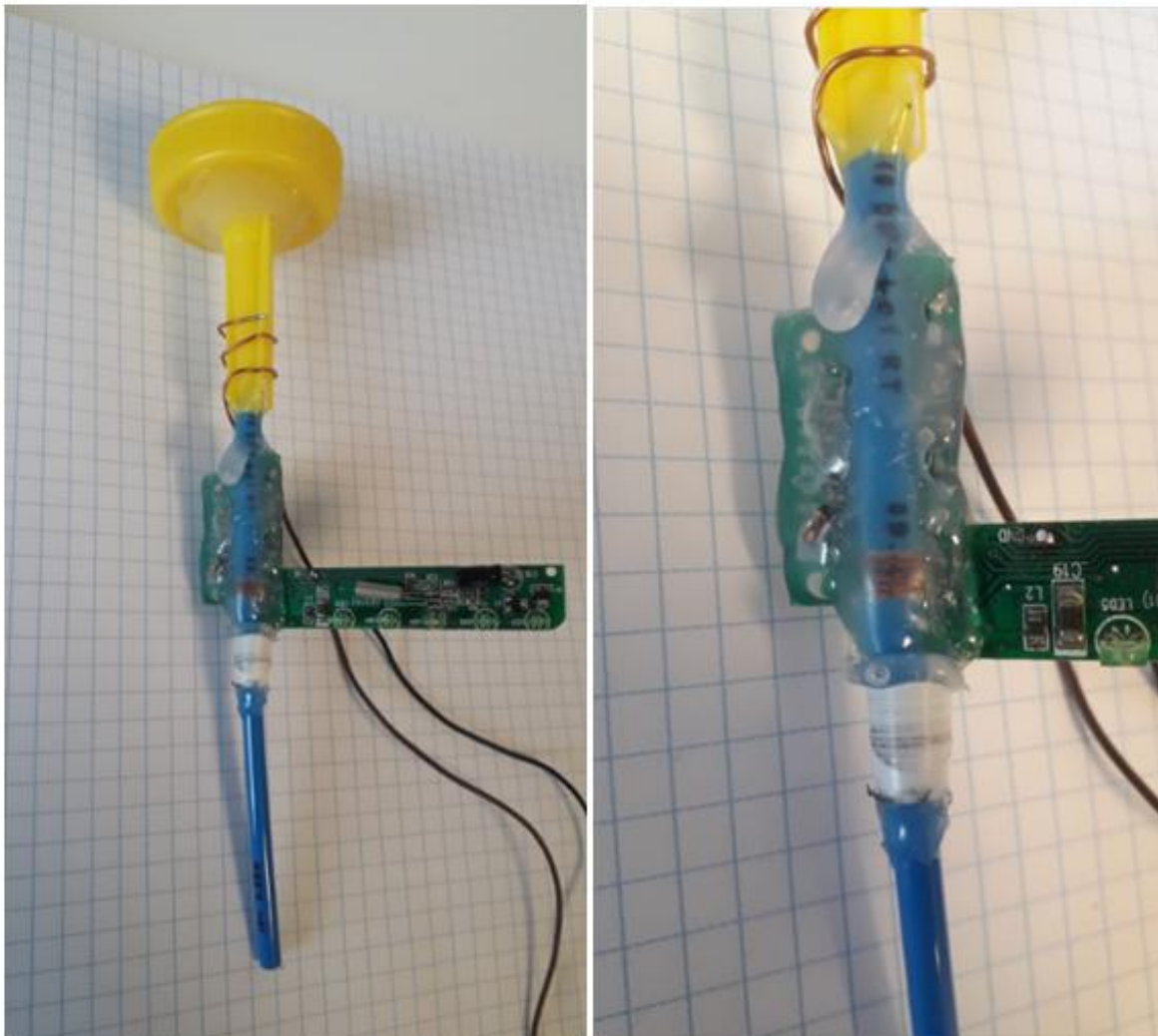


Figure 7. powder sensor

In fig. 8 is shown the audio signal acquired and filtered by the super heterodyne receiver, it is observed that there is an increase in the amplitude of the signal as the amount of powder introduced into the system increases.

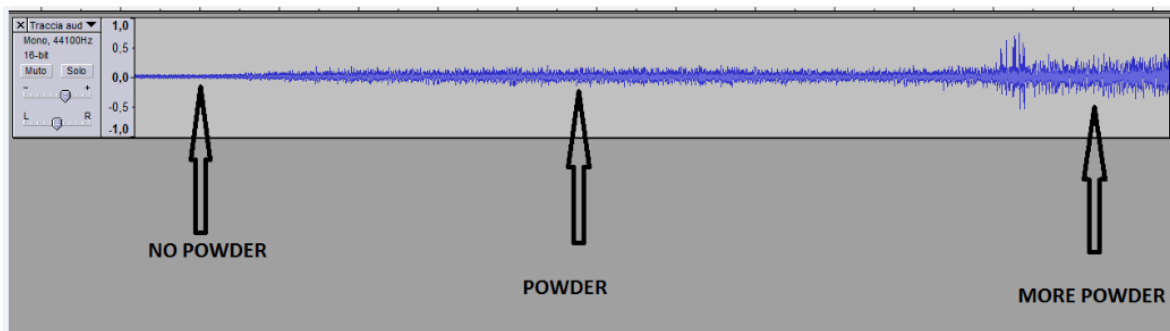


Figure 8. Data acquired by sampling the sound

In particular, in the FFT (Fast Fourier Transformer) graphs in fig. 9 you can see a correlation between frequency and maximum amplitude, it is observed that moment by moment only one frequency can be identified at the maximum amplitude peak of the detected signal.

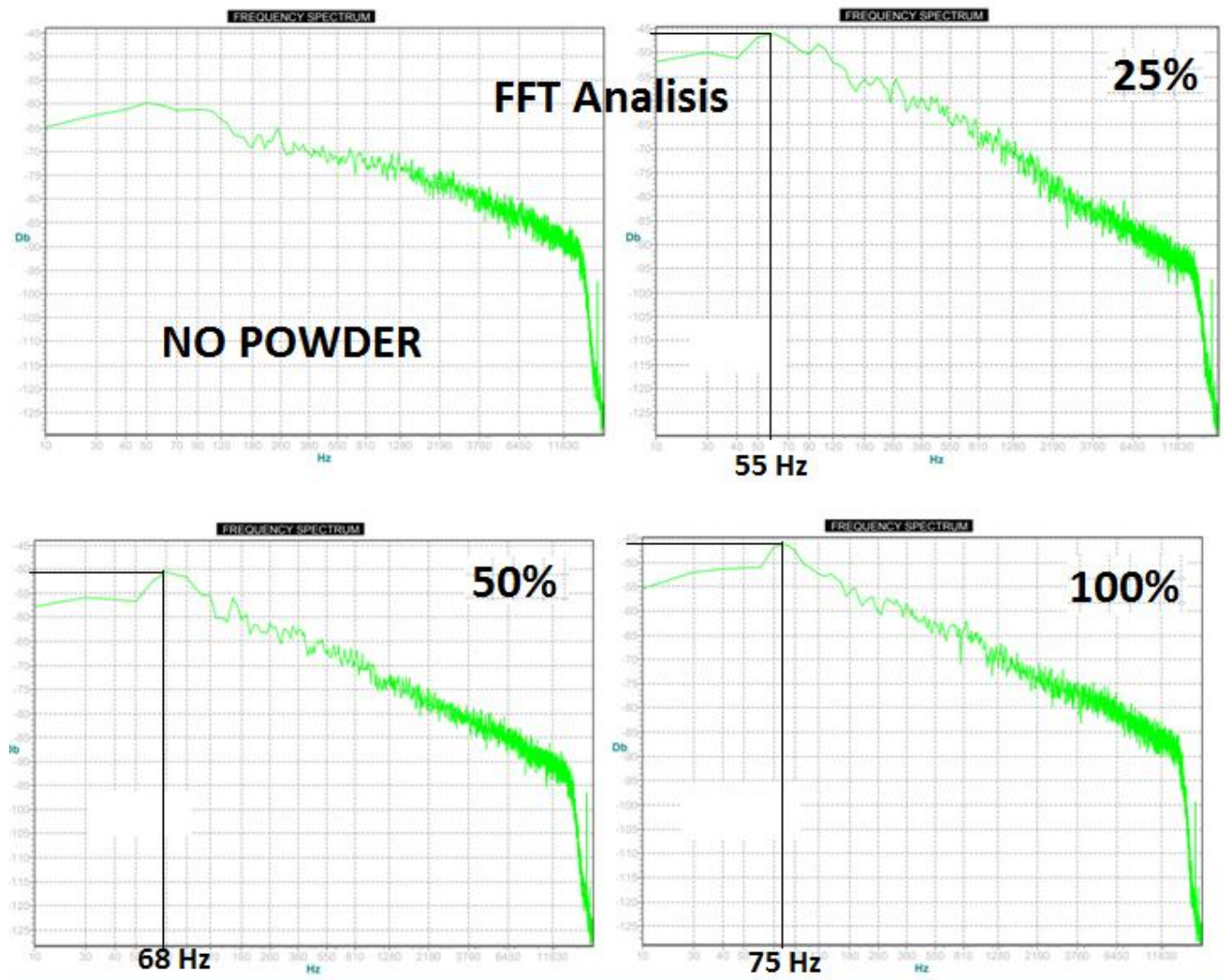


Figure 9. Spectral analysis of sound with different flow rate values

The whole system is controlled by an Arduino MKR 1010 module which performs the following tasks:

- PID controller
- communication with the outside world
- sampling of the signal acquired through the PWD sensor
- regulation of the powder flow by micro vibrator which is controlled via PWM.
- data storage



Figure 10. Arduino MKR WiFi 1010 controller

Communication with the outside world is implemented through 4 different channels:

- Bluetooth
- WiFi
- USB
- ETHERNET



Figure 11. Arduino MKR ETH shield

To have stability and robustness in the transmission of signals through the Ethernet shield, communication protocol **Modbus TCP** will be used.

The Ethernet shield is also equipped with a micro SD port for the storage of data recorded during system operation.

3.2.3 Optical setup for powder flow measurement

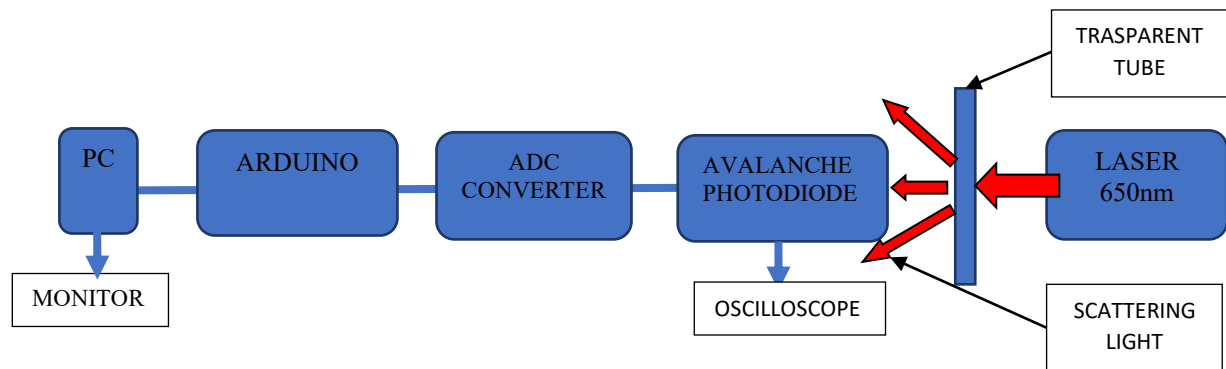
Another type of sensor that can be used to measure the flow of powder consists of a laser as transmitter unit and an avalanche photodiode as receiver unit. Between transmitter and receiver will be placed at a suitable distance a transparent tube at the wavelength of the laser (650 nm) in which the metal powder will flow.

The interaction between laser and powder will produce two main phenomena;

1. absorption of light by the powder
2. light scattering

The result of the interaction with the powder will therefore produce a dispersion of the light signal produced by the laser, consequently the receiver photodiode will detect only the part of the light that will not interact with the powder flow, therefore it will be possible to experimentally estimate the amount of powder that instantly flows inside the transparent tube.

The following Figure shows the simplified scheme.



Schematic diagram for laser-photodiode setup

Unfortunately, this method has not been tested because the covid-19 emergency slowed down considerably the supply of the components necessary for the construction of this equipment, but thanks to tests carried out with recycled materials it was possible to detect the variation produced by the interaction between laser powder and photodiode. However, such data are not available because the experiment was performed only to understand if it was possible to control the system with such a signal.

3.3 Conclusions

In this testing comprehended in August and September, different modalities of magnetic inductance principle for measuring of the powder flow were tested without a desired success, but tests carried out between February and July 2020, with the new sensor equipped with PLL (Phase-Locked Loop) have shown that the system is valid because it is sensitive enough to detect variations in powder flow.

In addition, some relevant conclusions have been reached:

- The relative quantity of the powder within the Argon flow is very small. It represents 0.04% of the overall volume of the flow. Hence, the influence of this quantity of the powder on the inductance seems to be very small, especially that Argon is a diamagnetic material with $\mu < 1$, which may diminish the influence of the powder. In fact, a significant change of frequency when the powder feeder is set on, contrasted to a non-conclusive variation with the powder mass rate change, indicate that probably the diamagnetic properties of Argon prevail over ferromagnetic influence of the small powder quantity, anyway this problem can be overcome using a technical expedient using a double oscillator that aims to eliminate the influence of argon gas instant by instant.
- The detected frequency variation was probably influenced by the environmental magnetic influence and vibrations, which can completely mask the variation caused by the powder flow. In other words, compared to the surrounding "noise", the magnetic response of the coil caused by

the powder flow is difficult to estimate, but with adequate shielding of the circuit it is possible to eliminate the disturbances produced by the external environment and make it more sensitive only to variations in the powder flow.

Some other tests were done with optical solutions that use a 3 mW LASER transmitter and an avalanche photodiode receiver that given good results, but the main problem was related to the tube in which the powder flows because to work fine that tube must be transparent to the wavelength of the LASER but when the powder passes through it, it attaches obscuring the path of the laser and creating diffraction of light so there is a decrease in the performance in a very short time.

In any case, to solve this problem it was decided to use a square section tempered glass tube in order to make the surface self-cleaning with the passage of the powder itself, this solution at the moment must still be tested.

4 Acknowledgement



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